

Traumatic Page kidney: a systematic review and patient report of bilateral Gerota's fascia release

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ABSTRACT

Background Page kidney is a rare cause of secondary hypertension resulting from renal parenchymal compression by subcapsular hematoma. Although trauma represents a well-established etiology, comprehensive analysis of traumatic Page kidney remains limited. This systematic review consolidates current evidence on trauma-induced Page kidney and presents the first reported bilateral patient managed with surgical decompression.

Methods After PRISMA 2020 guidelines, we systematically searched PubMed, Embase, and Web of Science databases through March 2023 using "Page kidney" AND "trauma." Two independent reviewers screened 119 abstracts, including 24 studies meeting eligibility criteria. Data extraction included demographics, trauma mechanism, clinical presentation, imaging findings, treatment modalities, and outcomes. Additionally, we present a novel bilateral case requiring surgical intervention.

Results Twenty-four patients with traumatic Page kidney were analyzed (median age 24.5 years, 95.8% male). Blunt trauma predominated (50%), followed by falls (25%) and motor vehicle collisions (25%). Hypertension occurred in 87.5% of patients, with a median time to diagnosis of 15 days (range 0 year to 9 years). Left kidney involvement was most common (41.7%), with 16.7% occurring in renal allografts. Surgical decompression was performed in 37.5% of patients, percutaneous drainage in 25%, and nephrectomy in 20.8%. No mortality was reported. Our bilateral case involved a 54-year-old man who developed anuria after a motor vehicle collision, successfully treated with bilateral Gerota's fascia release.

Conclusions Traumatic Page kidney predominantly affects young men after blunt trauma, with frequent diagnostic delays. Surgical decompression achieves excellent outcomes when conservative management fails. Bilateral involvement is exceptionally rare but treatable with aggressive surgical intervention. Trauma surgeons should maintain clinical suspicion for Page kidney in patients with post-traumatic hypertension or unexplained renal dysfunction.

Level of evidence Level IV, therapeutic/care management

BACKGROUND

Page kidney is a rare but potentially serious secondary cause of hypertension initially described by Irvine Page in 1939 from mouse models.¹ The condition results from a subcapsular fluid collection that exerts enough compression to activate the renin-angiotensin-aldosterone system, leading

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Page kidney is a rare cause of secondary hypertension from renal parenchymal compression, with bilateral traumatic cases representing extreme rarity and unclear optimal management strategies.

WHAT THIS STUDY ADDS

⇒ This is the first reported bilateral traumatic Page kidney successfully managed with bilateral Gerota's fascia release, and the largest systematic review identifying surgical decompression in 37.5% of 24 patients with zero mortality despite frequent diagnostic delays (median 15 days, range 0 year to 9 years).

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Bilateral Page kidney should be recognized as a urological emergency requiring immediate surgical intervention, and trauma surgeons should maintain heightened suspicion for this diagnosis in patients with post-traumatic hypertension or unexplained renal dysfunction, even years after injury.

to severe hypertension and renal dysfunction.² Although its pathophysiology resembles the experimental Goldblatt kidney model, Page kidney occurs through extrinsic rather than intrinsic vascular compression.³

The epidemiology of Page kidney has shifted considerably since it was first described. Early patient series identified trauma as the primary cause, particularly sports injuries and motor vehicle collisions.^{4–6} However, more recent reports frequently cite iatrogenic causes, such as complications from renal biopsy and spontaneous hematomas due to anticoagulation, especially in patients who had transplants.^{7–9} This shift is attributed to improved protective gear for risky activities and the increasing use of kidney-related interventional procedures.

Bilateral Page kidney is an extremely rare form of this already uncommon condition. The simultaneous development of compressive perinephric collections in both kidneys creates a unique clinical challenge, as the usual compensatory mechanisms of the opposite kidney are lost.¹⁰ The limited literature suggests that bilateral involvement may occur in cases of anticoagulation, systemic vasculitis, or severe bilateral trauma, but a thorough analysis of these cases has been lacking.^{11–13}

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Recognition of Page kidney in trauma patients presents specific challenges because of the potential for delayed symptoms and the initial focus on more immediate, life-threatening injuries. Patients may develop symptoms days to years after the initial trauma, and diagnosing this condition requires a high level of suspicion in patients who present with new-onset hypertension and a history of renal trauma.^{14–16} The bilateral form adds further diagnostic complexity and also creates an urgent need for intervention to prevent permanent kidney damage.

Current management strategies for Page kidney range from medical management to surgical decompression, with treatment choice influenced by factors such as timing of presentation, severity of symptoms, and underlying comorbidities.^{17–19} However, suggested guidelines for bilateral patients are lacking due to the extreme rarity of this presentation. Most management recommendations are based on case reports and small case series, which limits the ability to establish standardized treatment protocols.

To our knowledge, this study reports the first known case of bilateral traumatic Page kidney successfully treated with bilateral surgical decompression, along with a systematic review of all patients with trauma-related Page kidney documented in the literature. Our aims were to describe the epidemiology and clinical features of traumatic Page kidney, evaluate treatment approaches and outcomes, and offer suggested recommendations for managing this rare but potentially serious condition.

METHODS

Systematic literature review

We conducted a comprehensive systematic literature review following PRISMA 2020 guidelines.²⁰ The search was performed on March 21, 2023, using PubMed, Embase, and Web of Science databases. Search terms included “Page kidney” OR “Page phenomenon” AND “trauma.” No language restrictions or publication date limits were applied to ensure the complete capture of all relevant cases. The full search strategy is available in online supplemental digital content 1 (SDC1).

Two independent reviewers screened titles and abstracts based on predefined inclusion and exclusion criteria. Inclusion criteria included any case report or case series describing Page kidney secondary to trauma in human subjects with enough clinical detail for data extraction. Exclusion criteria covered iatrogenic Page kidney without related trauma, animal studies, review articles without original case data, and cases lacking sufficient clinical information for analysis. Blunt renal trauma was defined as a non-penetrating injury to the kidney caused by external mechanical forces, such as, but not limited to, motor vehicle collisions, falls from height, contact sports, physical assaults, and crush injuries, as opposed to iatrogenic injury from medical procedures. See online SDC2: Reviewer Worksheet for further details.

Full-text articles were reviewed to select eligible studies, with disagreements resolved through consensus discussion. Data extraction was conducted using a standardized form that captured patient demographics, injury mechanisms, clinical presentation, diagnostic findings, treatment methods, and clinical outcomes (online SDC 3: Data Dictionary). A follow-up literature search was performed in April 2025 to identify any additional cases published since the initial review. All data analysis was conducted using R statistical software, and the data and analysis are available on the following repository: github.com/anthonyomorada/page-kidney-trauma-sr/.

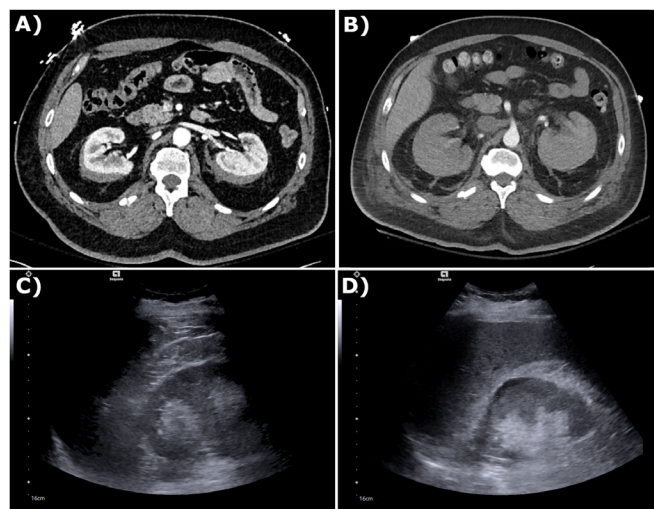


Figure 1 Imaging of traumatic bilateral Page kidney. (A) Initial CT scan showing bilateral perinephric hematomas without active extravasation. (B) Repeat CT angiography on hospital day 4 demonstrating enlarging bilateral zone II perinephric hematomas with mass effect on both kidneys. (C/D) Postoperative imaging confirming resolution of the perinephric collections and restored renal anatomy.

RESULTS

Case report

A patient in their 50s presented to our level I trauma center after a motor vehicle collision as an unrestrained driver. The motor vehicle collision occurred at approximately 35 mph with airbag deployment. Initial trauma evaluation revealed a 5 cm scalp laceration over the left parietal region and right-sided rib fractures involving ribs 9 through 12. CT imaging demonstrated bilateral perinephric hematomas without evidence of active extravasation (figure 1A). The patient's initial vital signs were stable, with a blood pressure of 142/88 mm Hg, a heart rate of 92 beats per minute, and an oxygen saturation of 98% on room air. Laboratory studies revealed a hemoglobin level of 12.4 g/dL and a creatinine level of 1.1 mg/dL, both within normal limits.

The patient was admitted to the trauma service for serial monitoring and pain management. Their tertiary survey remained unremarkable, and they initially had stable clinical parameters. However, on hospital day 4, the clinical team observed progressive deterioration in renal function with a rising serum creatinine to 2.3 mg/dL, along with worsening hypertension with systolic pressures consistently over 180 mm Hg despite antihypertensive treatment. At the same time, their hemoglobin dropped to 9.8 g/dL without evidence of ongoing bleeding from their known injuries.

Given these concerning clinical changes, repeat CT angiography was performed, which showed enlarging bilateral zone II perinephric hematomas with evidence of mass effect on both kidneys (figure 1B). The imaging indicated compression of the renal parenchyma on both sides, consistent with the development of bilateral Page kidney phenomenon. The patient's urine output also decreased to less than 0.5 mL/kg/hour despite adequate fluid resuscitation.

Recognizing the risk of permanent kidney damage from ongoing compression, the decision was made to proceed with surgery. The patient underwent an exploratory laparotomy with bilateral exploration, bilateral Gerota's fascia release, and evacuation of the zone II hematomas. Surgical findings showed significant bilateral retroperitoneal swelling with organized hematomas

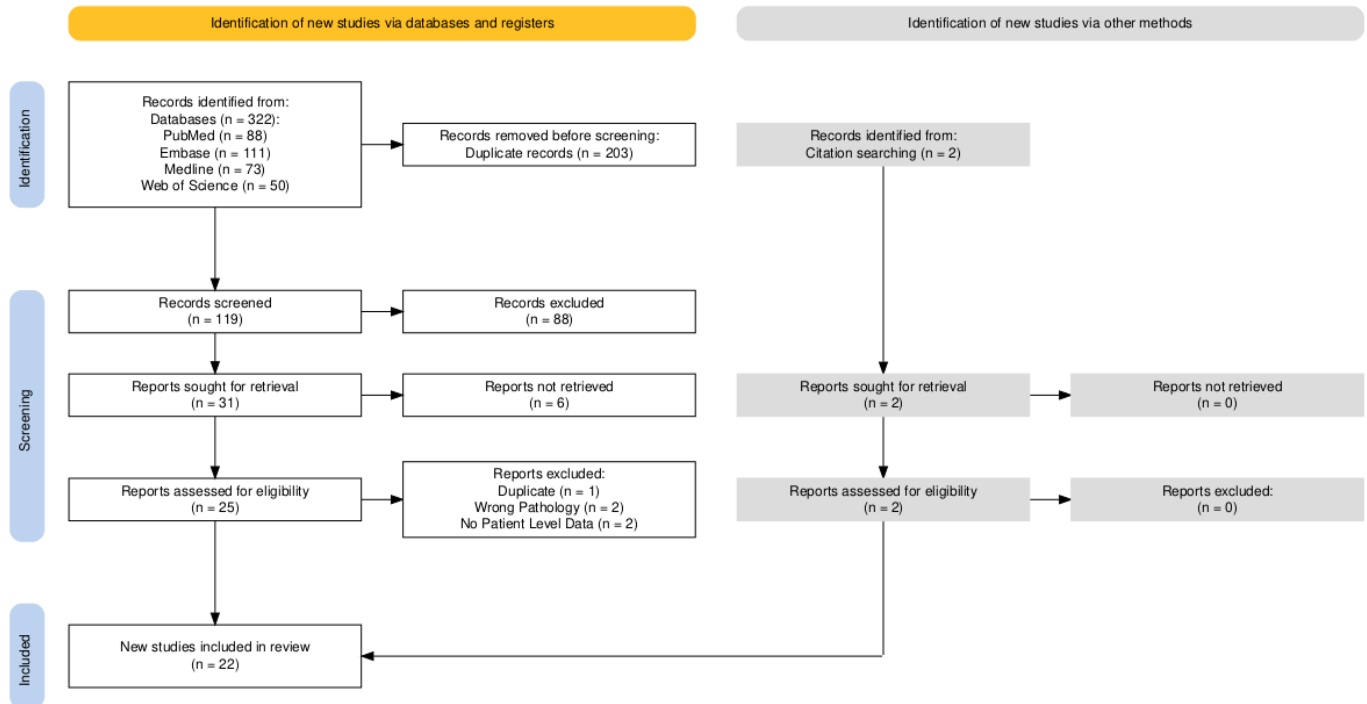


Figure 2 PRISMA flow diagram for systematic review of traumatic Page kidney. Flow diagram depicting the identification, screening, eligibility assessment, and inclusion of studies according to PRISMA 2020 guidelines. The initial database search across PubMed, Embase, MEDLINE, and Web of Science yielded 119 articles. After duplicate removal using Rayyan software, 98 unique articles underwent title and abstract screening by two independent reviewers. Thirty-one articles met initial screening criteria and proceeded to full-text review. After application of inclusion and exclusion criteria, 24 articles reporting individual cases of traumatic Page kidney were included in the final analysis. Reasons for exclusion during full-text review included: non-traumatic etiology (n=4), insufficient clinical detail for data extraction (n=2), duplicate patient reports (n=1), and inability to access the full text despite contact attempts (n=0). A follow-up literature search conducted in April 2025 using identical methodology identified two additional eligible cases published between March 2023 and April 2025, bringing the total included cases to 26. All included studies underwent data extraction using a standardized form, with quality assessment based on completeness of clinical information and follow-up data. The flow diagram demonstrates the systematic and transparent approach used to identify and select relevant literature for this comprehensive review of patients with traumatic Page kidney.

around both kidneys. Two laparotomy pads were placed in each perinephric space to prevent reaccumulation, and an ABThera wound vacuum-assisted closure device was used, considering the temporary nature of the initial closure. The patient returned to the operating room 48 hours later for a planned second-look procedure. The laparotomy pads were removed, a thorough washout was performed, and primary fascial closure was achieved without tension. Both kidneys appeared well perfused with normal color and consistency after decompression.

Postoperatively, the patient was managed in the intensive care unit and was successfully extubated on postoperative day 3. Despite the successful surgical decompression, the patient's renal function continued to decline, requiring the placement of a temporary hemodialysis catheter and the initiation of renal

replacement therapy on postoperative day 5. The patient's blood pressure normalized after the surgical intervention, needing only minimal antihypertensive support.

During the next few days, the patient showed gradual clinical improvement with stable hemoglobin levels and resolution of their hypertension. They were able to advance their diet and switch to oral pain medications. However, their kidney function recovery was delayed, and they remained on dialysis at the time of discharge to a rehabilitation facility on postoperative day 9, resulting in a total hospital stay of 14 days.

At the 2-week follow-up in the outpatient clinic, the patient showed significant improvement with complete resolution of their hypertension and normalization of their renal function. They had been successfully weaned off dialysis and exhibited

Table 1 Special populations in traumatic Page kidney

Population	N (%)	Mean age (years)	Unique considerations
Renal transplant recipients	4 (16.7%)	58.5	Superficial location, immunosuppression
Bilateral Page kidney	2 (8.3%)	35	Risk of acute renal failure
Pediatric patients (<18 years)	8 (33.3%)	15.3	Growth considerations, organ preservation priority
Delayed diagnosis (>30 days)	10 (41.7%)	31	Higher risk of irreversible changes
Transplant recipients, n (%)	2 (15.4%)	2	(20.0%) >0.99*

*Fisher's exact test.

Table 2 Patient characteristics by time to diagnosis

Characteristic	Acute (≤ 30 days) N=13	Delayed (>30 days) N=10	P value
Mean age (years)	38.1	31	0.45*
Male gender, n (%)	13 (100%)	9 (90%)	0.43†
Nephrectomy rate, n (%)	1 (7.7%)	4 (40.0%)	0.09†
Bilateral involvement, n	1 (7.7%)	0 (0%)	$>0.99†$
Transplant recipients, n (%)	2 (15.4%)	2 (20.0%)	$>0.99†$

Delayed diagnosis (>30 days) associated with increased nephrectomy rate, approaching statistical significance ($p = 0.09$).
 *Mann-Whitney U test.
 †Fisher's exact test.

adequate urine output. Follow-up imaging confirmed the resolution of the perinephric hematomas with normal renal parenchymal enhancement on both sides (figure 1C,D). The patient reported feeling well, returning to their baseline functional status, with no residual symptoms related to their traumatic injuries or subsequent Page kidney phenomenon.

To our knowledge, this case is the first reported instance of bilateral traumatic Page kidney treated with bilateral surgical decompression, emphasizing both the rarity of this condition and the potential for excellent outcomes when recognized early and managed appropriately.

SYSTEMATIC REVIEW

Literature search and study selection

The systematic literature search conducted on March 21, 2023, identified 119 articles across PubMed, Embase, Medline, and Web of Science databases. After removing duplicates and screening abstracts, 31 articles proceeded to full-text review. After the application of inclusion criteria, 20 articles met the eligibility requirements, representing 21 individual patients with trauma-related Page kidney (see online supplemental table 1). A follow-up search in April 2025 found two additional case reports published after the initial review, which were added to the final analysis. The PRISMA flow diagram showing the selection process is presented in figure 2. The publication timeline spans 55 years from 1969 to 2024, with 62.5% of cases published between 2010 and 2024, indicating increased awareness of this condition in recent years.^{4–19 21–25} Including our case of bilateral Page kidneys, we collected 23 publications with 24 patients.

Demographics

Analysis of the 24 patients included showed a median age at presentation of 24.5 years (IQR 17 years to 54 years; range 9 years to 89 years), with a mean age of 34.2 ± 21.7 years (see online supplemental table 1). There was a strong male predominance, with 23 men (95.8%) and one woman (4.2%).

Past medical history data showed that four patients (16.7%) had previous renal transplants, making it one of the most

common comorbidities. Other notable comorbidities included diabetes mellitus in three patients (12.5%), hypertension in two patients (8.3%), and congenital kidney anomalies in two patients (8.3%). The high number of renal transplant recipients suggests that this group may have an increased risk of developing Page kidney after trauma, as detailed further in our analysis of special populations (table 1).

Clinical presentation and diagnosis

The time from initial trauma to Page kidney diagnosis showed significant variability, with a median of 15 days (IQR 5 days to 243 days). However, the average time was 367 ± 813 days, indicating several patients with very delayed presentations. The range spanned from immediate diagnosis on the trauma day to 9 years afterwards, highlighting the need to stay vigilant for Page kidney long after the injury.

Mechanisms of injury were classified as blunt trauma in 12 patients (50%), falls in 6 patients (25%), and motor vehicle collisions in 6 patients (25%). The predominance of blunt trauma mechanisms aligns with the common injury patterns seen in renal trauma.

The clinical presentation was marked by hypertension in 21 patients (87.5%), making it the most common feature. Other symptoms included flank pain in 8 patients (33.3%), hematuria in 7 patients (29.2%), abdominal pain in 4 patients (16.7%), and back pain in 2 patients (8.3%). Notably, one patient (4.2%) was asymptomatic at diagnosis, discovered incidentally during imaging for other reasons. The widespread presence of hypertension emphasizes its importance as a diagnostic indicator for Page kidney (table 2).

Imaging and laboratory findings

Urinalysis results were available for 14 of 24 patients (58.3%). Of those tested, 8 patients (57.1%) had hematuria, and 6 patients (42.9%) had normal urinalysis. The lack of hematuria in a significant number of patients suggests that a normal urinalysis does not exclude the diagnosis of Page kidney.

Table 3 Treatment outcomes by management strategy

Treatment strategy	N (%)	Mean age (years)	Median time to diagnosis (IQR)*	Organ preservation
Conservative management	3 (12.50)	43 8 days	(3 days to 3.0 months)	Yes
Percutaneous intervention	6 (25.00)	38.2 22 days	(1 day to 2.0 months)	Yes
Surgical decompression	7 (29.20)	38.9 15 days	(6 days to 1.0 year)	Yes
Nephrectomy	5 (20.80)	26.6 2.0 years	(0 days to 3.0 years)	No
Other Interventions†	3 (12.50)	19.3 4 days	(1 day to 6 days)	Yes

*Time from trauma to Page kidney diagnosis.

†Includes urologic stenting and vascular interventions. All patients survived through the available follow-up period.

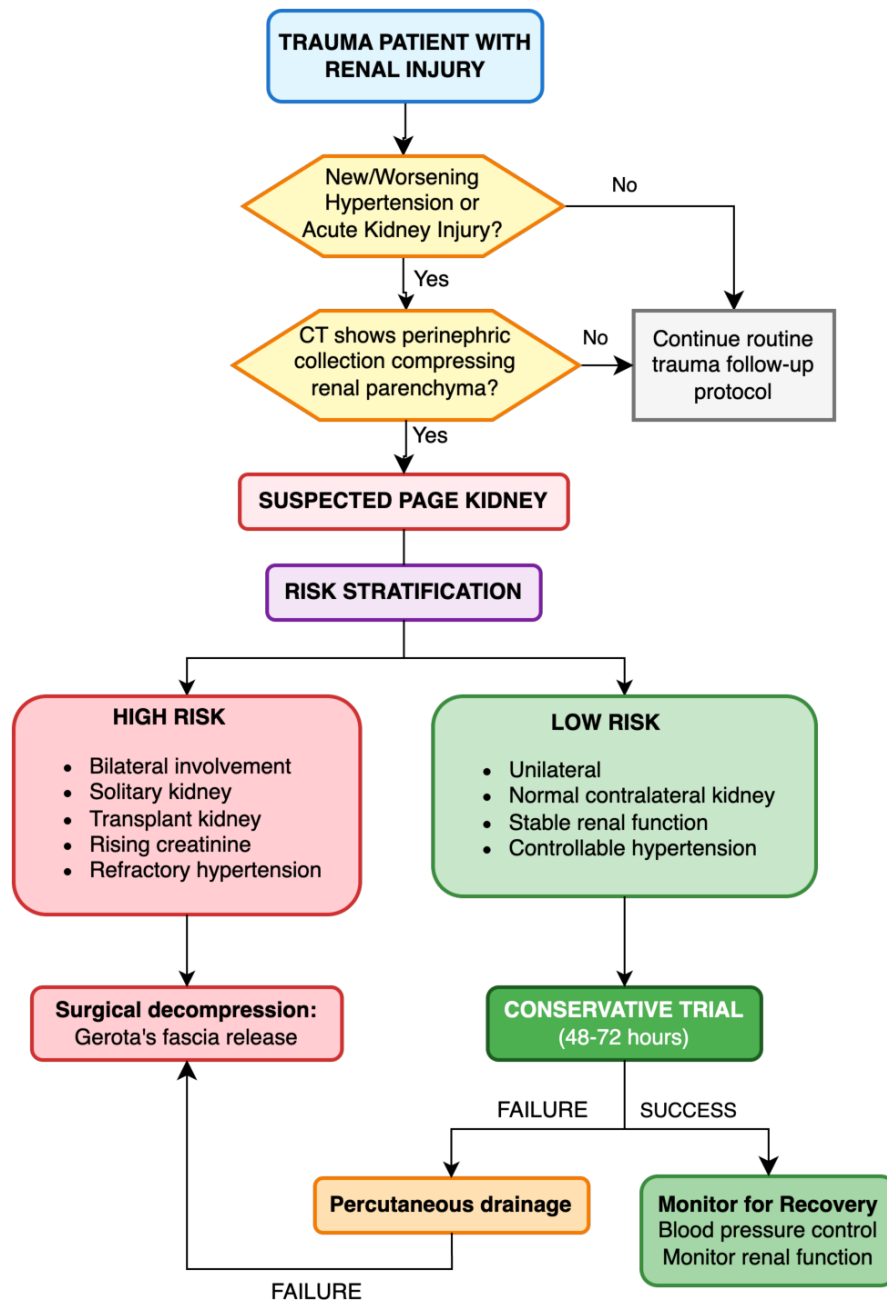


Figure 3 Evidence-based clinical decision algorithm for traumatic Page kidney. This algorithm incorporates risk stratification based on systematic review findings, with high-risk factors including bilateral involvement, solitary kidney, transplant status, rising creatinine, and refractory hypertension. Conservative management success rates and escalation pathways are based on treatment patterns observed in the literature. The algorithm emphasizes organ preservation and ensuring timely intervention for high-risk patients.

Documentation of injury severity using the American Association for the Surgery of Trauma kidney injury scale was limited, with grades reported in only three patients (12.5%). Among these, one patient each had grade II, IV, and V injuries. The limited injury grading data reflect inconsistent reporting standards in the literature and hinder stratification by injury severity.

The size of perinephric fluid collections at presentation was documented in 13 patients (54.2%), with a median size of 11 cm (IQR 7.4 cm to 12 cm) and an average of 10.7 ± 4 cm. The smallest collection measured 4 cm, and the largest was 18 cm. Hematoma was the most common type of fluid collection, accounting for 21 patients (87.5%), followed by 1 patient each of urinoma (4.2%) and lymphocele (4.2%).

Laterality and anatomic distribution

Analysis of laterality revealed left kidney involvement in 10 patients (41.7%), right kidney involvement in 8 patients (33.3%), bilateral involvement in 2 patients (8.3%), and allograft involvement in 4 patients (16.7%). The predominance of left-sided involvement may reflect anatomic differences, as the left kidney is positioned slightly higher and has a longer renal vein, which could affect susceptibility to compressive forces. The significant proportion of allograft patients highlights the vulnerability of transplanted kidneys to the Page phenomenon, likely due to their superficial location and altered anatomic relationships.

Treatment modalities and outcomes

Treatment approaches varied significantly, reflecting changes in management strategies and individual patient factors (table 3). Surgical decompression was the most common intervention, performed in 9 patients (37.5%), including in 7 as the sole treatment and in 2 in combination with percutaneous drainage. Percutaneous drainage was used in 6 patients (25.0%), nephrectomy in 5 patients (20.8%), medical management in 3 patients (12.5%), urological interventions in 2 patients (8.3%), and vascular interventions in 2 patients (8.3%). One patient received both percutaneous and vascular interventions.

Treatment options were significantly affected by the time to diagnosis (table 3). Patients who needed nephrectomy had the longest median time to diagnosis (2.0 years, IQR 0 days to 3.0 years), and those treated with kidney-preserving methods were diagnosed within days to months (median 8 days to 22 days). This timing emphasizes the importance of early detection in preventing permanent kidney damage.

Follow-up data were available for 20 patients (83.3%), with an average follow-up of 1.1 years (median 0.38 years; range 0.01 year to 9 years). Notably, there were no deaths among the 24 patients. Although specific data on blood pressure control and long-term renal function were inconsistently reported, the absence of mortality and the resolution of symptoms in most patients indicate favorable short-term to medium-term outcomes with proper intervention.

Time-to-diagnosis analysis

Analysis of patients divided by time to diagnosis showed essential differences in treatment outcomes (table 2). Patients diagnosed within 30 days of trauma ($n=13$) had significantly lower rates of nephrectomy compared with those with delayed diagnosis during 30 days ($n=10$): 7.7% vs 40.0% ($p=0.09$, Fisher's exact test). This indicates a 5.2-fold higher risk of organ loss with delayed recognition. The group with delayed diagnosis also showed lower overall kidney preservation rates (60.0% vs 92.3%), highlighting the importance of timely diagnosis.

Special populations

Several patient subgroups exhibited distinct clinical patterns (table 1). Renal transplant recipients accounted for 16.7% of patients ($n=4$), with an average age of 58.5 years and a tendency towards surgical decompression, likely due to the vulnerability of superficially placed allografts to external pressure. Patients with bilateral Page kidney ($n=2$, 8.3%) were surgical emergencies requiring immediate intervention to prevent acute renal failure. Pediatric patients (<18 years, $n=8$, 33.3%) were managed with organ-preserving strategies when possible, highlighting the importance of conserving renal mass for future growth and development.

Comparison with historical data

The patients with bilateral Page kidney identified in our review, including our index patient, form an extremely rare subgroup. Only two bilateral cases were found in the systematic review, making up 8.3% of the cohort. This rarity highlights the importance of our case report, to our knowledge, as the first documented instance of bilateral traumatic Page kidney successfully treated with bilateral Gerota's fascia release.

This systematic review represents the most comprehensive analysis to date of patients with traumatic Page kidney, aggregating evidence from diverse case reports and series spanning several decades. Although rare, the Page kidney phenomenon is

a notable pitfall in trauma care—it can transform what initially seems like a successfully managed renal injury into a potentially life-threatening hypertensive complication later on. Our review provides several important insights for clinicians.

DISCUSSION

To our knowledge, our index case represents the first documented bilateral traumatic Page kidney successfully managed with bilateral surgical decompression, demonstrating that even the most complex presentations can achieve complete functional recovery with prompt recognition and appropriate intervention. The rapid progression from initial trauma to anuria within 4 days, followed by complete renal recovery after surgical decompression, underscores both the urgency and reversibility of this condition when properly managed.

The systematic review reveals critical patterns that inform clinical decision-making for traumatic Page kidney. The overwhelming male predominance (95.8%) and young median age (24.5 years) reflect trauma exposure patterns, but more importantly suggest that clinicians should maintain heightened suspicion in this demographic. The wide variability in time to diagnosis—ranging from immediate recognition to 9 years post-trauma—represents a fundamental clinical challenge requiring sustained vigilance for Page kidney in any patient with new-onset hypertension after abdominal or flank trauma.^{4 14}

Treatment paradigms have included organ preservation, defined as keeping the native kidney, with surgical decompression now representing the most common intervention (37.5% of patients), and nephrectomy has decreased to only 20.8% of patients. This dichotomy reflects improved understanding that Page kidney results from mechanical compression rather than intrinsic renal abnormality, making organ-preserving approaches both feasible and preferred.^{6 17} All patients undergoing surgical decompression in our series achieved resolution of hypertension with preserved renal function, supporting this therapeutic approach.

In our systematic review of traumatic Page kidney, all four reported nephrectomy patients involved a clearly established requirement for nephrectomy, as validated by functional renal imaging, intraoperative findings, or histopathological evidence of irreversible parenchymal injury. Emergency hemorrhage control was the primary indication in Murray's patient, and Mullins and both Sokhal patients demonstrated non-functioning kidneys via nuclear medicine imaging prior to nephrectomy.²⁶ Of particular note, Sokhal's second patient illustrates the natural progression of delayed Page kidney: laparoscopic decompression initially resulted in transient blood pressure improvement, yet follow-up imaging at 3 months showed development of a non-functioning kidney. Histopathology confirmed replacement of renal parenchyma with necrotic fibrocollagenous tissue and marked glomerular and tubular compression, underscoring the permanent nature of chronic compression-related damage. This example highlights that, in chronic patients with established fibrosis, organ-preserving approaches may not succeed, ultimately necessitating nephrectomy. Collectively, these findings indicate that the link between delayed diagnosis and nephrectomy is driven by documented irreversible renal damage rather than practice variation alone, though the precise temporal threshold after which salvage is impossible remains unclear.

Renal transplant recipients represent a uniquely vulnerable population, comprising 16.7% of our traumatic patients despite their small proportion of the general population. This disproportionate representation likely reflects the superficial

location of transplanted kidneys, altered vascular anatomy, and potential bleeding tendencies related to immunosuppression.^{8,9} Even minor trauma—including a 15-pound cat jumping on the abdomen—can precipitate Page kidney in this population, necessitating specific surveillance protocols and lower thresholds for imaging after any abdominal trauma.

Bilateral involvement, though rare (8.3% of patients), represents a true urological emergency requiring immediate intervention to prevent irreversible renal failure. Unlike unilateral patients where contralateral compensation maintains overall renal function, bilateral Page kidney eliminates this safety mechanism and mandates aggressive management.¹⁰ Our patient demonstrates that bilateral surgical decompression can be safely performed with excellent outcomes, providing a treatment paradigm for future cases of this critical presentation.

The absence of hematuria in 50% of patients with documented urinalysis challenges traditional screening approaches and emphasizes that normal urinalysis cannot exclude Page kidney. This finding shifts diagnostic emphasis from urinary markers to hemodynamic parameters, particularly new-onset or refractory hypertension after trauma.^{5,10}

Based on these findings, we recommend risk stratification incorporating bilateral involvement, solitary kidneys, transplant status, and diagnostic delay as high-risk factors requiring urgent evaluation and intervention. Early CT angiography should be considered for patients with refractory hypertension or acute kidney injury after trauma, particularly when perinephric collections are suspected. Surgical decompression should be strongly considered for patients demonstrating significant renal parenchymal compression, especially in high-risk populations (figure 3).

This study has several important limitations. First, although our bilateral patient improved after surgical decompression, we cannot conclusively determine that spontaneous resolution would not have occurred without intervention. Second, this review is subject to significant publication bias inherent in case report literature. Only patients with clinically apparent Page kidney are included; however, many trauma patients likely experience subcapsular hematomas with transient renal compression that either do not manifest as Page kidney or resolve spontaneously without diagnosis. The actual number of at-risk patients and the natural history of untreated renal compression remain unknown. Finally, variability in outcome reporting across case reports limited our ability to conduct robust comparative analyses of treatment efficacy.

Future research should focus on developing standardized diagnostic protocols and treatment algorithms through prospective multicenter registries. Given the extreme rarity of bilateral cases, international collaborative databases will be essential for meaningful analysis of this critical subset. Trauma protocols should incorporate Page kidney awareness as a potential delayed complication, with specific emphasis on high-risk populations including transplant recipients and contact sport athletes.^{7,11}

Our successful management of bilateral traumatic Page kidney demonstrates the potential for excellent outcomes with prompt recognition and appropriate surgical intervention, expanding the therapeutic literature for this rare but serious condition. The combination of systematic review findings and this novel case presentation provides comprehensive suggestions supporting an aggressive, organ-preserving approach to traumatic Page kidney management, with the potential to prevent irreversible complications and maintain long-term renal function.^{5,6}

CONCLUSIONS

Traumatic Page kidney represents a rare but reversible cause of secondary hypertension after blunt renal trauma that requires heightened clinical awareness for optimal outcomes. Our systematic review demonstrates that delayed diagnosis significantly impacts treatment success, with patients diagnosed beyond 30 days experiencing substantially higher nephrectomy rates. To our knowledge, this study presents the first reported bilateral traumatic Page kidney successfully managed with bilateral surgical decompression, demonstrating that even complex presentations requiring temporary dialysis can achieve complete functional recovery with prompt intervention. Renal transplant recipients represent a disproportionately vulnerable population, comprising 16.7% of traumatic patients, with even minor trauma capable of precipitating Page kidney in this group. Organ-preserving surgical approaches offer superior outcomes compared with historical nephrectomy rates, and trauma protocols should incorporate Page kidney awareness as a potential delayed complication, particularly in high-risk populations including transplant recipients and patients with solitary kidneys.

Contributors AOM: guarantor, primary writer, responsible for initial data collection, participated in two physician review states in the study. AKL: secondary writer, assisted in two physician review states in the study, responsible for edits and submission as corresponding author. AD: assisted in conception of the article. RL: supervision of the project, assisted in draft reviews.

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